Regression

Shuvongkar Sarkar

Regression

- In the previous class, we covered the concept of "correlation," which helps us measure the relationship between two variables, denoted as "x" and "y”.

- A positive correlation implies that as "x" increases, "y" also increases, or vice versa.

- However, determining which variable is influencing the other is challenging.

Regression

- Relationship between two or more variables
- Cause and effect relationship
- Casual variables vs Affected variables
- Causal variables as independent variables; affected variables as dependent variables

Regression

$$Dependent\ variable = Y$$

$$Independent\ variable = X$$

- Regression refers to the cause-and-effect relationship between two or more variables, where affected variables are dependent variables and causal variables are independent variables.

- This relationship is mathematically expressed, highlighting the impact of independent variables on dependent variables.

Regression

- There is a positive relationship between income and expenditure, i.e. an increase in income increases expenditures.

- As increase in income causes an increase in expenditures

- “Income” as independent variable (X) and “Expenditures” as dependent variable (Y).

Correlation vs Regression

Correlation	Regression
1. Correlation is a statistical measure that determines the association between two variables.	1. Regression describe how to numerically relate an independent variable to the dependent variable.
2. There is no dependent variable and independent variable.	2. Must be one dependent variable and one independent variable.
3. To represent the linear relationship between variables.	3. To represent the cause and effect relationship between variables.

Types of Regression

- On the basis of variables (two types)



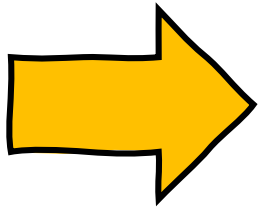
Types of Regression

- On the basis of variables (two types)



Types of Regression

- On the basis of variables (two types)



Simple regression

- One dependent vs one independent

Multiple regression

- One dependent vs multiple independent

Simple regression

- One dependent variable vs One independent variable
- Expressed this relationship as a mathematical form
- The mathematical form is,

$$Y_i = \alpha + \beta X_i + \epsilon_i \quad ; i = 1,2,3, \dots, n$$

Here,

$Y_i = Dependent\ variable$

$X_i = Independent\ variable$

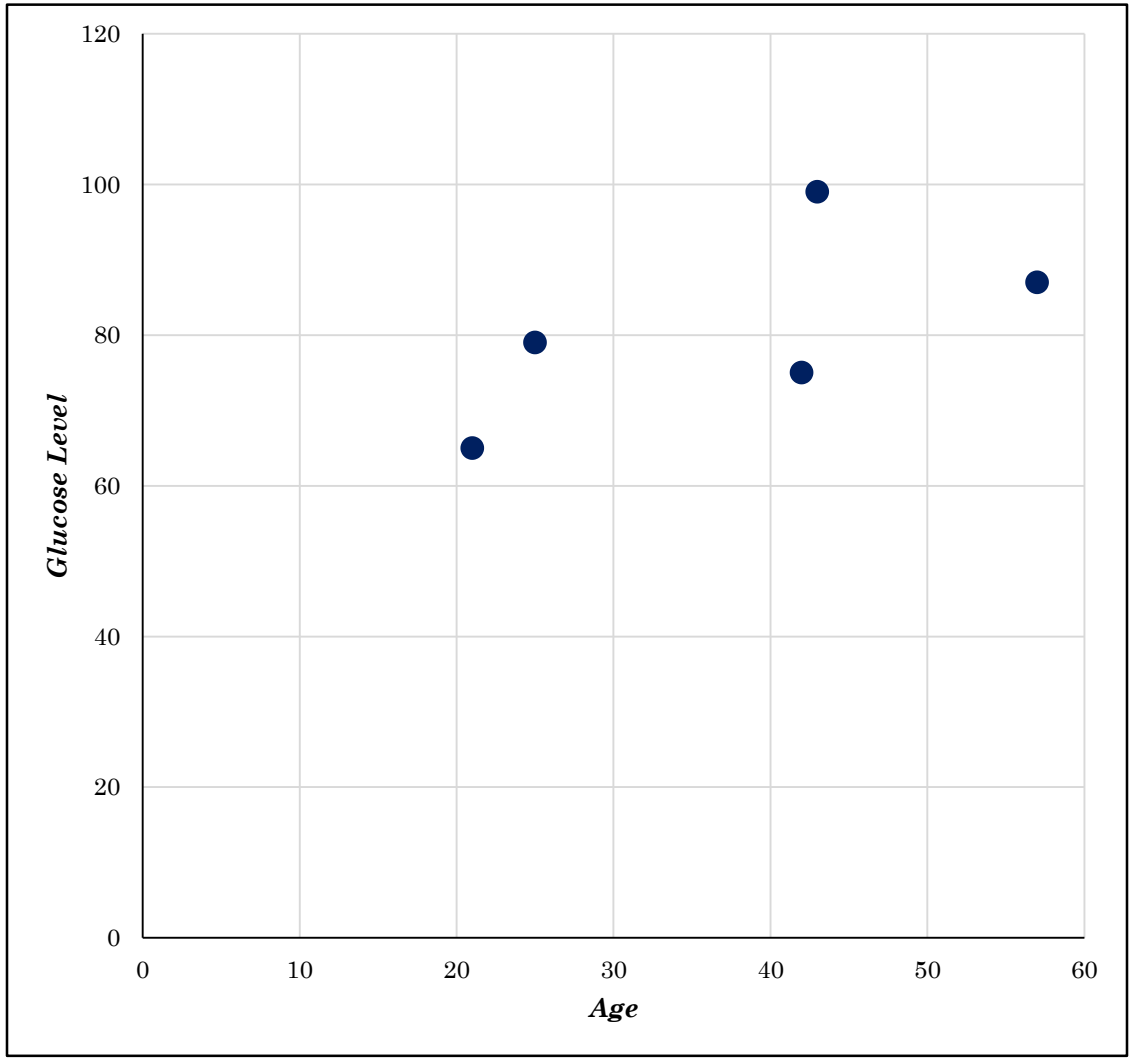
$\alpha = Intercept\ coefficient$

$\beta = Slope\ coefficient$

$\epsilon_i = Error\ term$

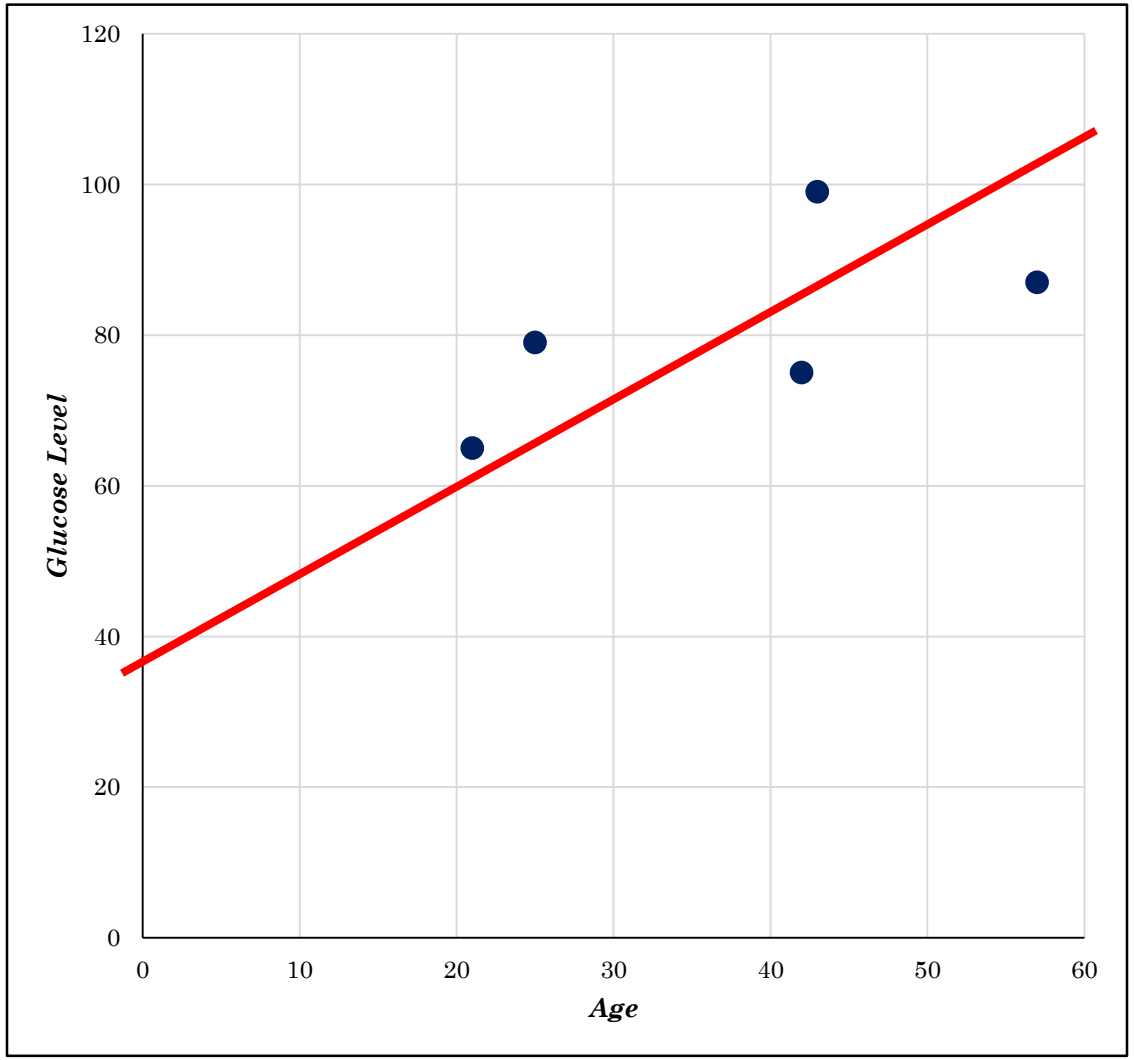
Multiple regression: $Y = \alpha + \beta_1 X_1 + \cdots + \beta_k X_k + \epsilon$

Simple regression



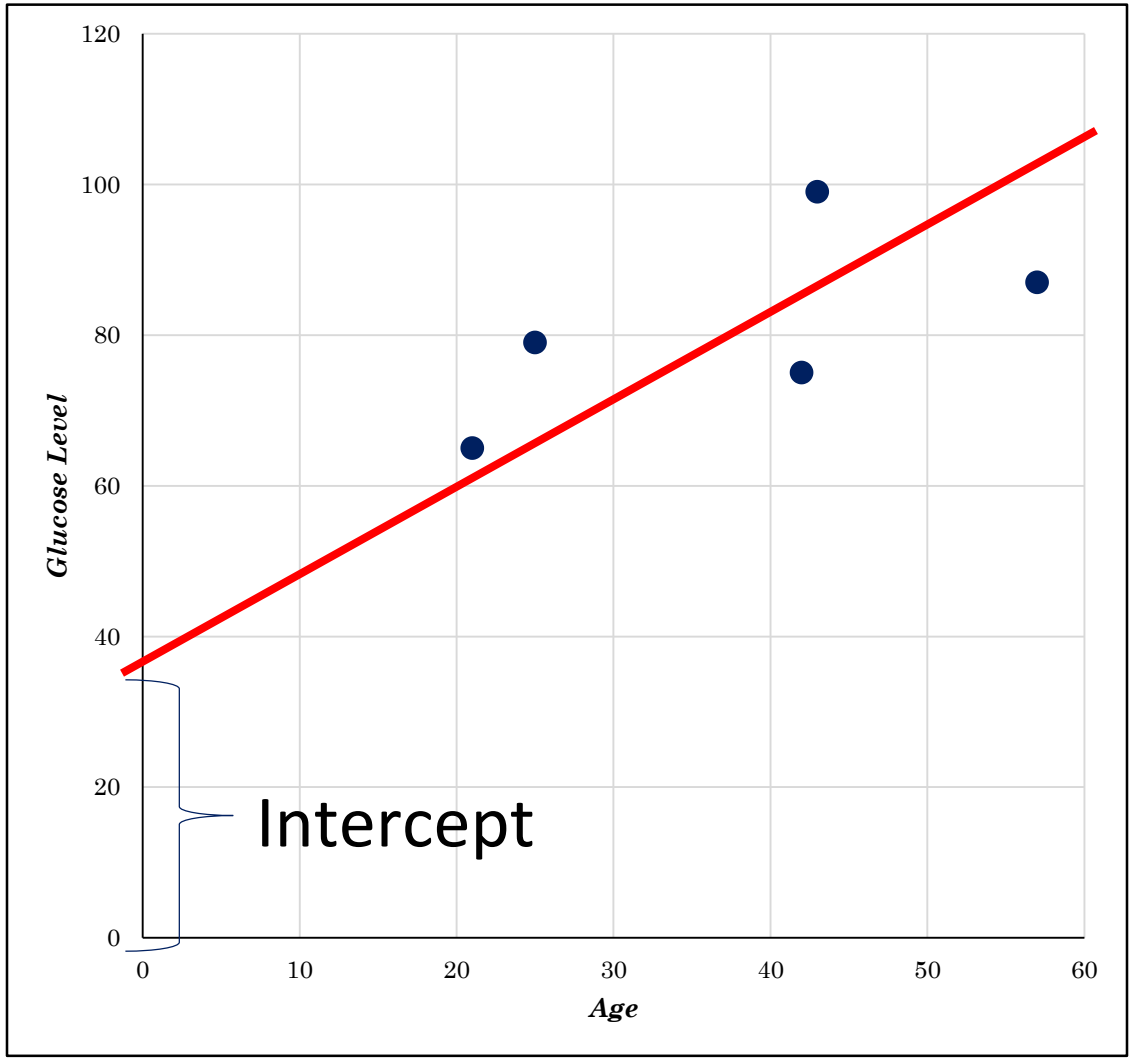
Simple regression

$$Y_i = \alpha + \beta X_i + \epsilon_i$$



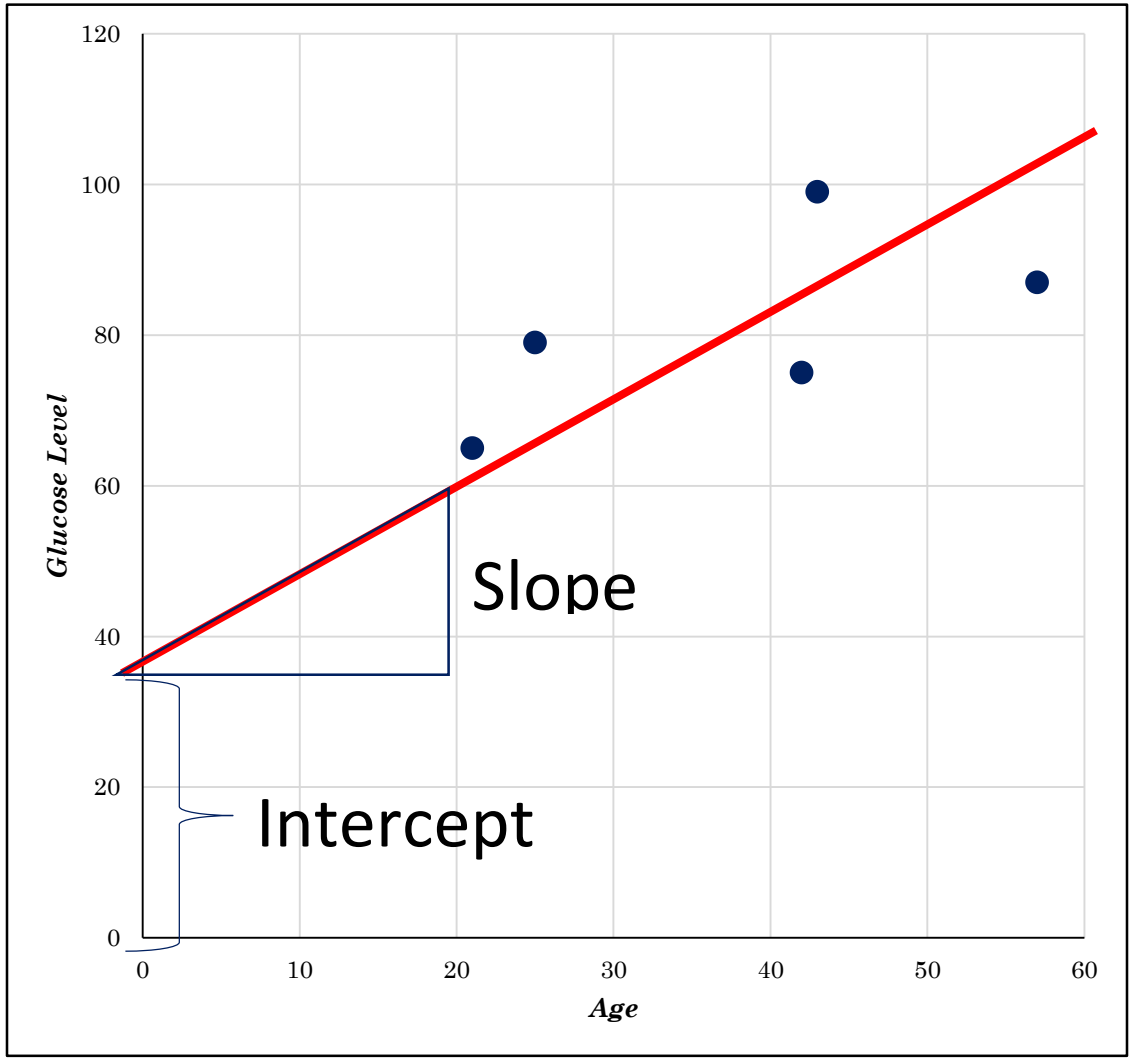
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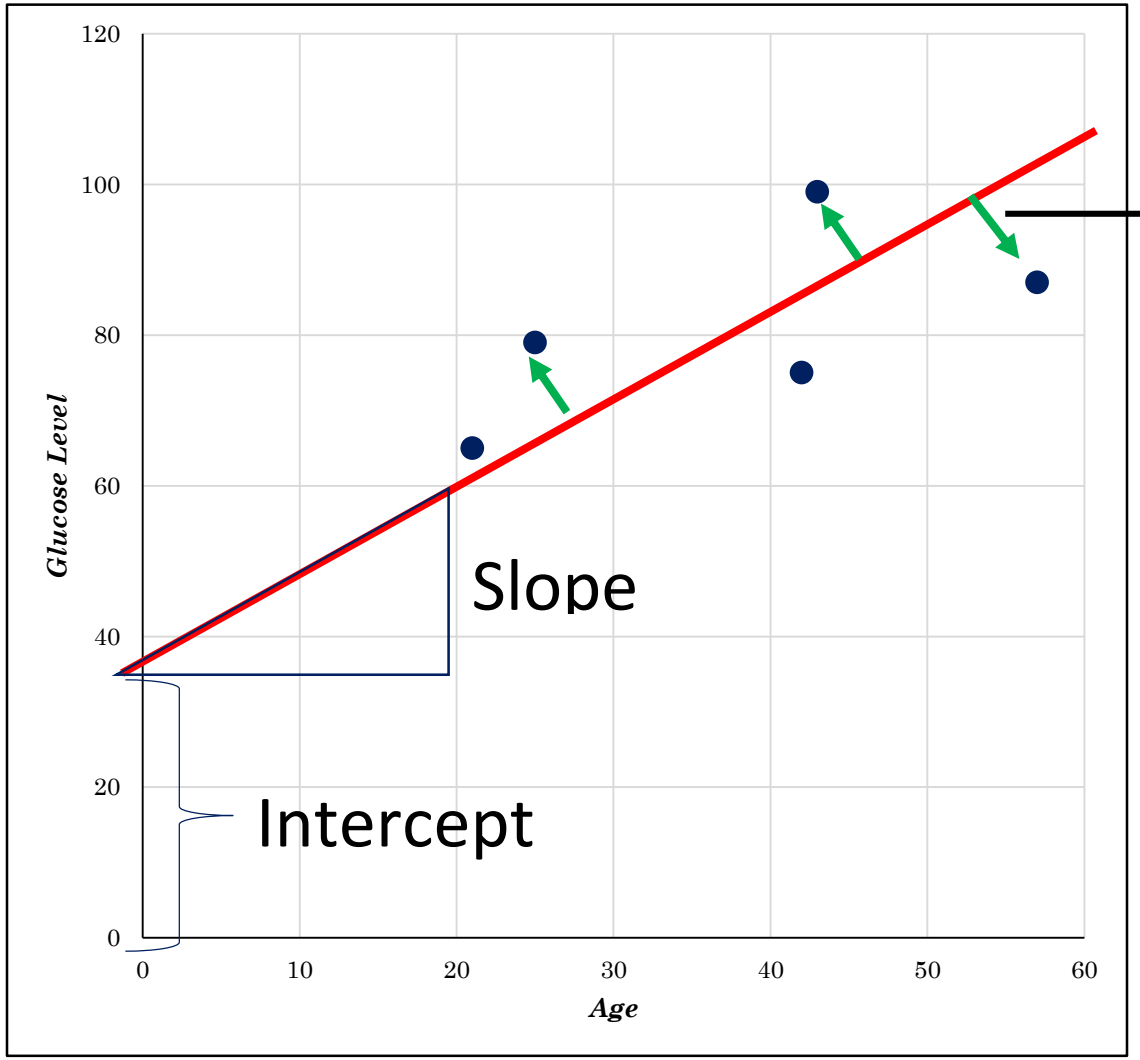
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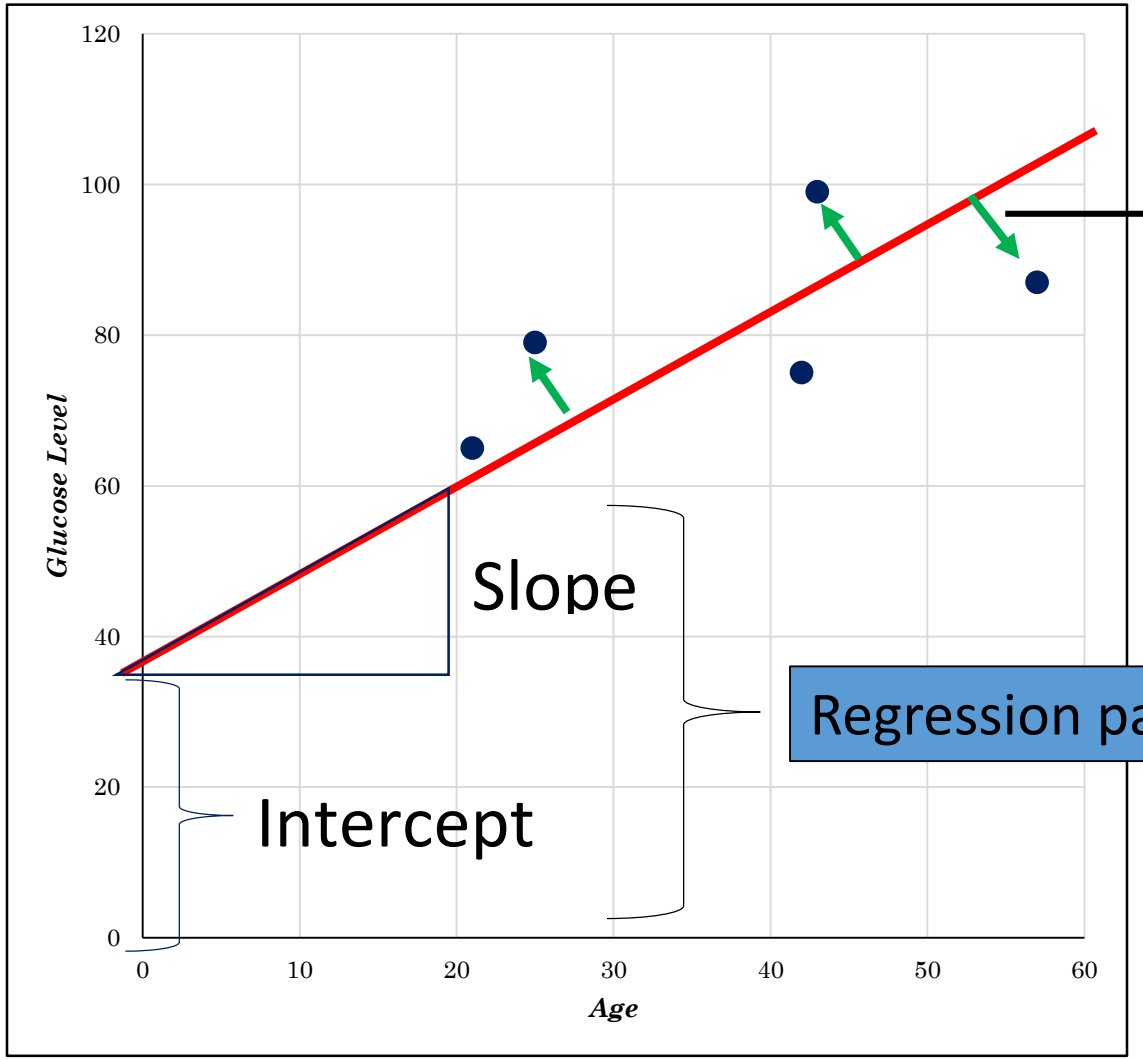
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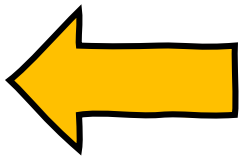


Error term

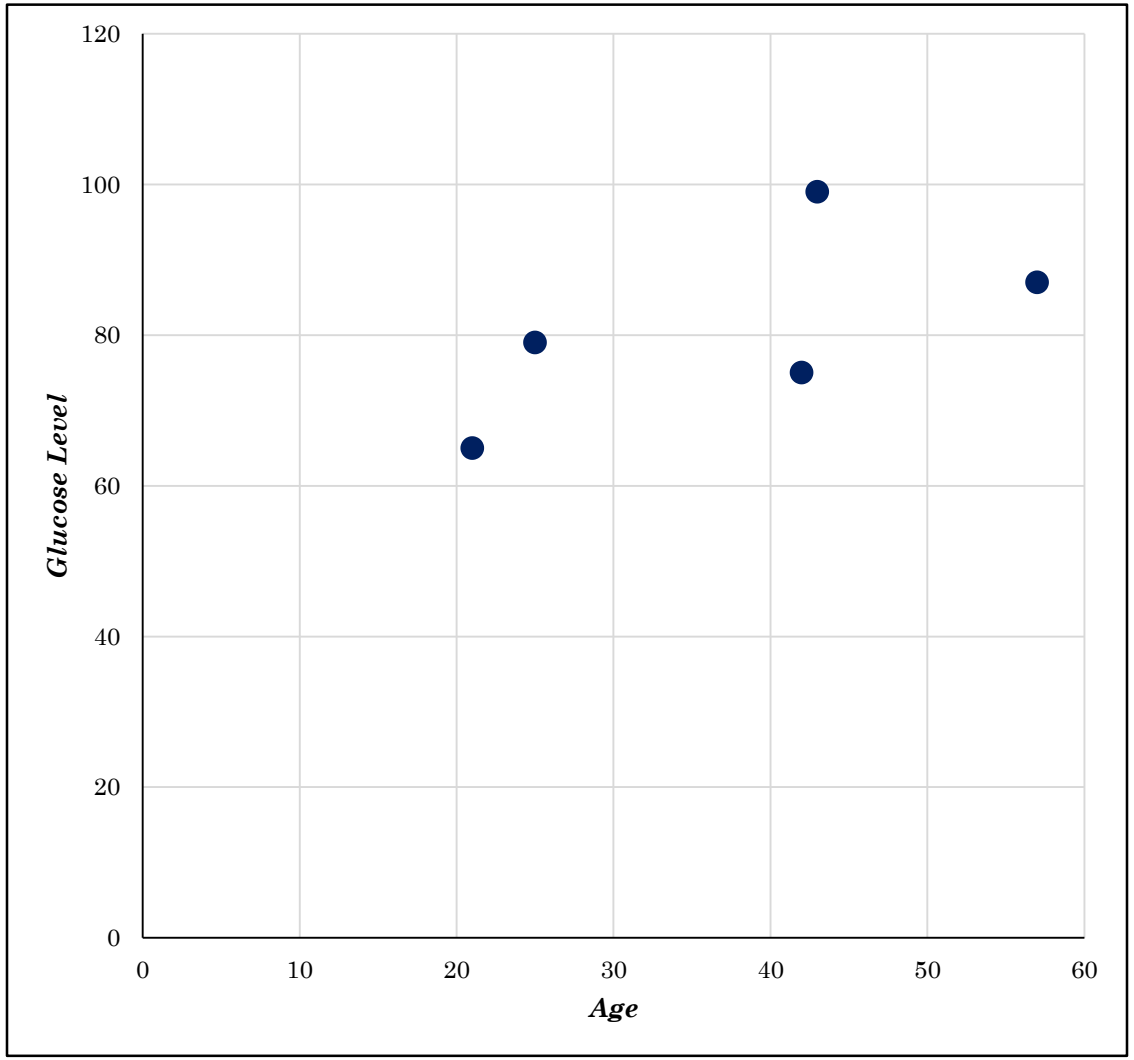
Regression parameter

Estimating parameters

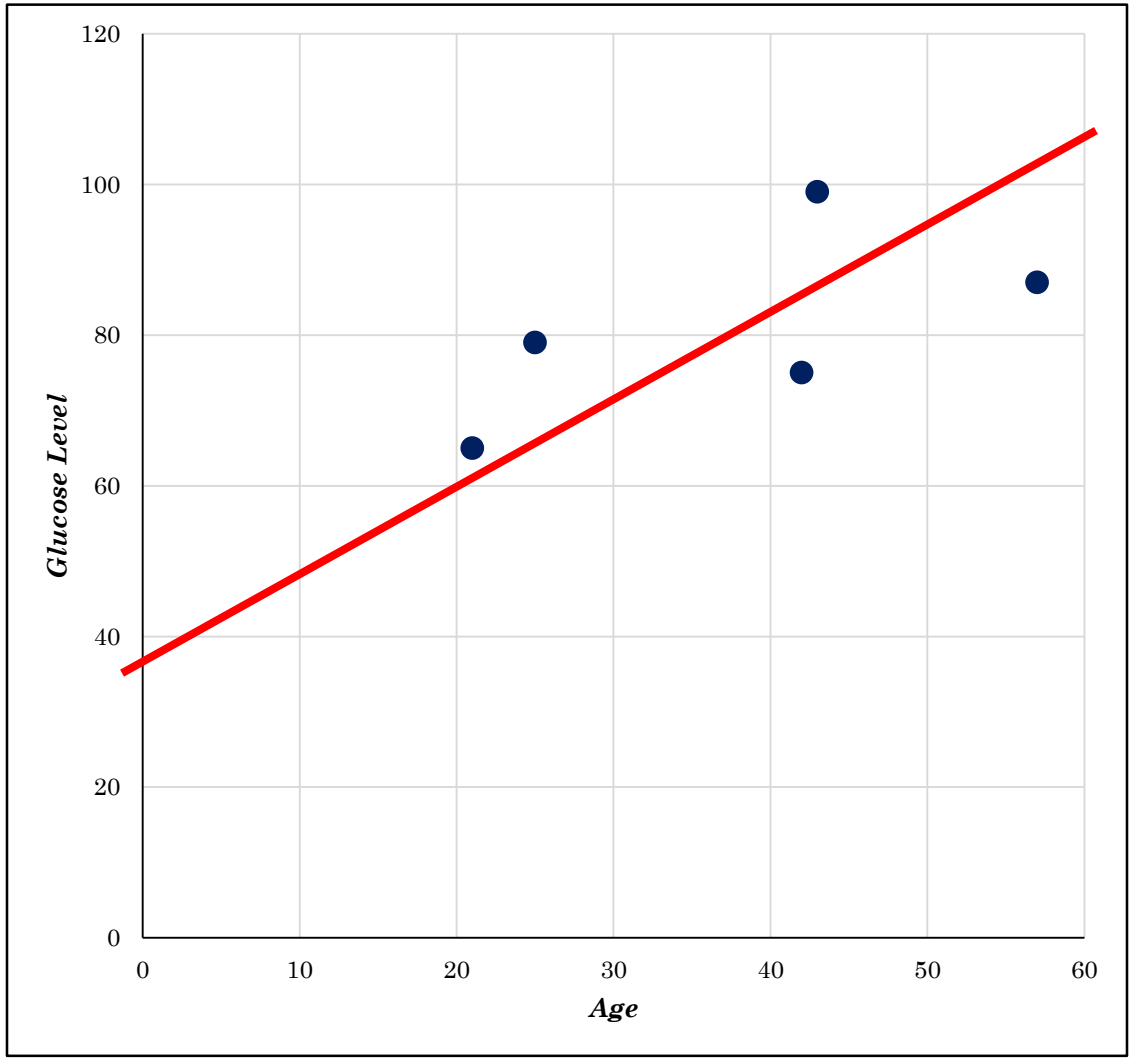
- Estimating regression parameters:

1. Least Square Method 
2. Graphical Method

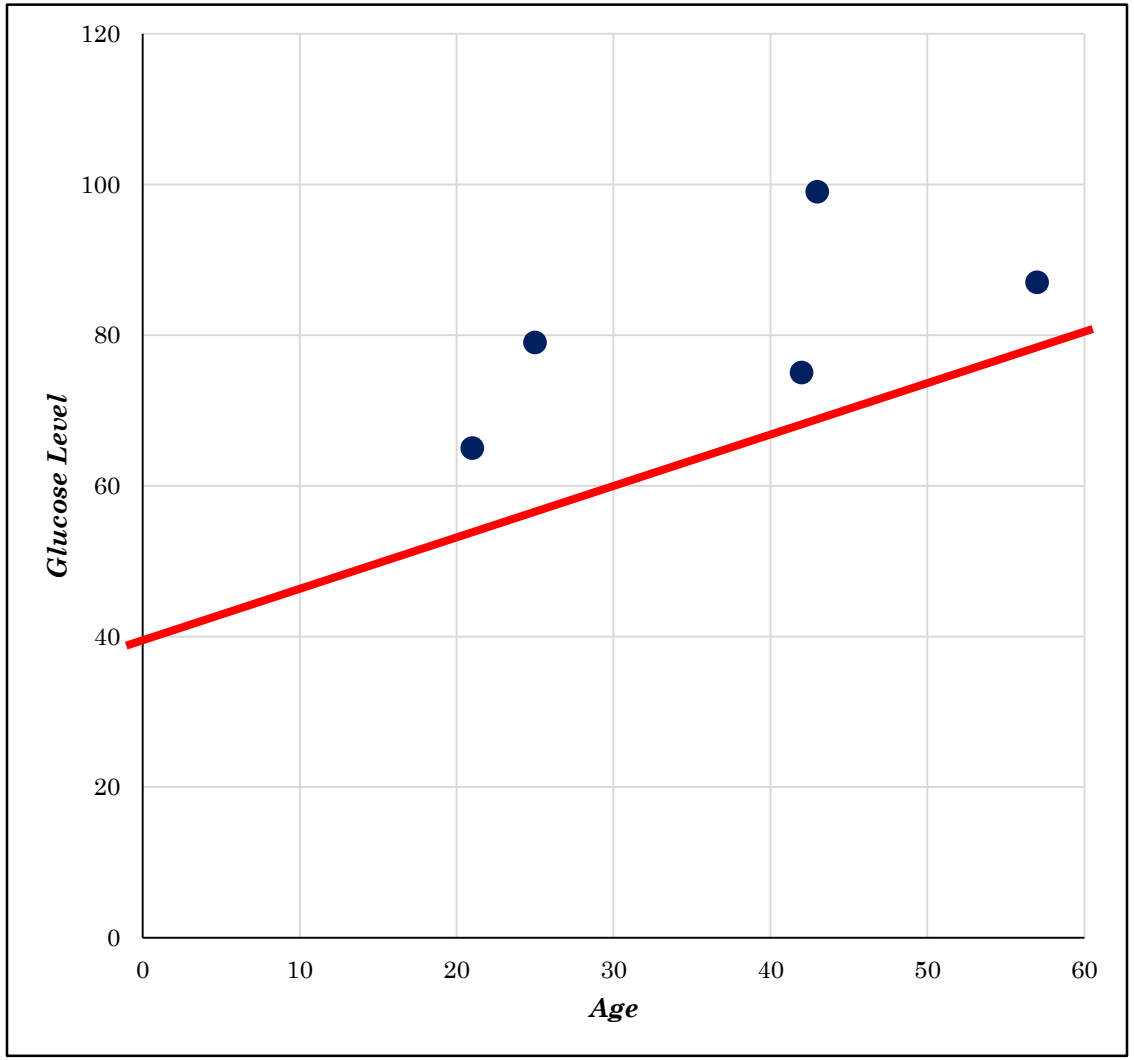
Estimating parameters



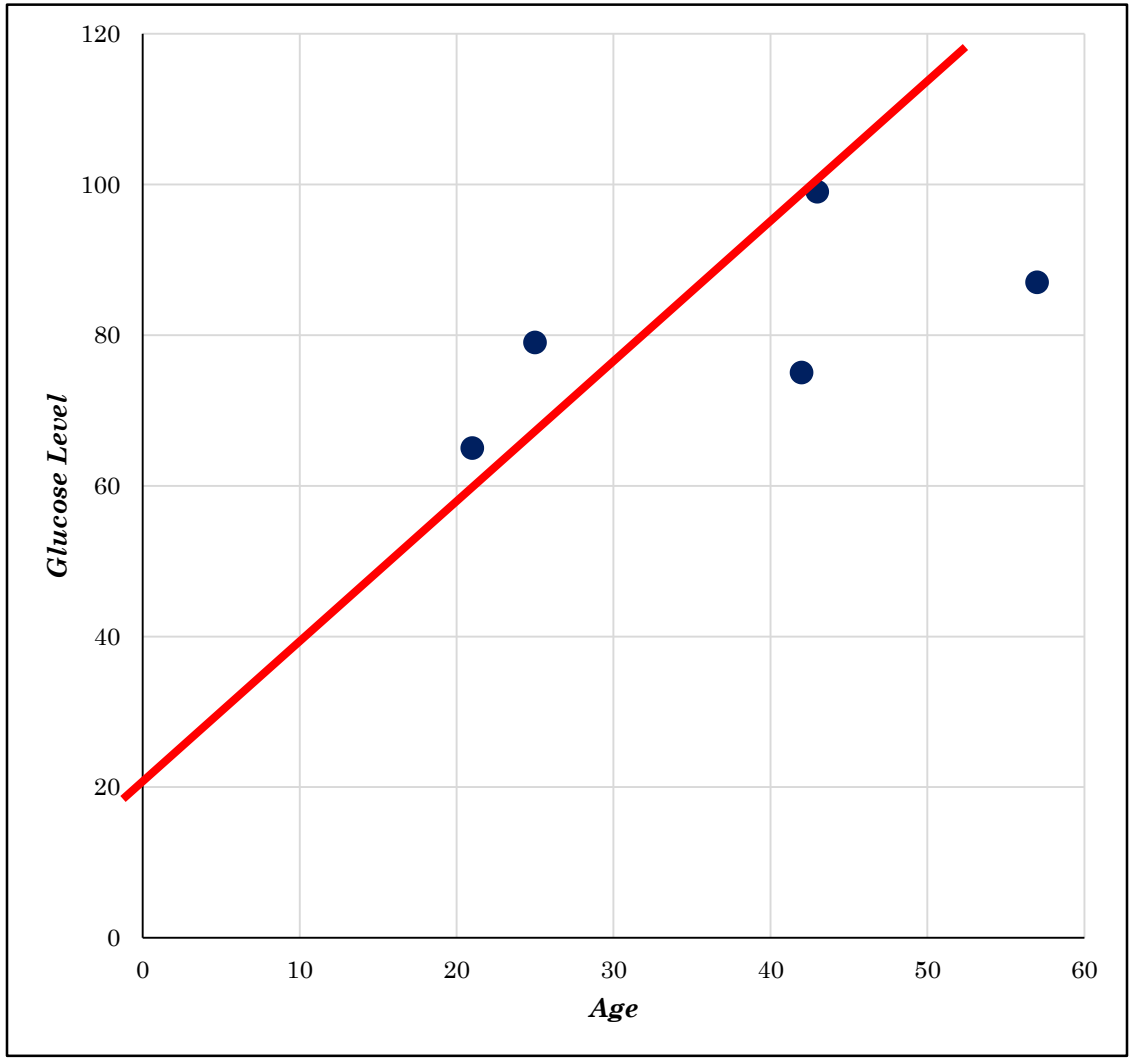
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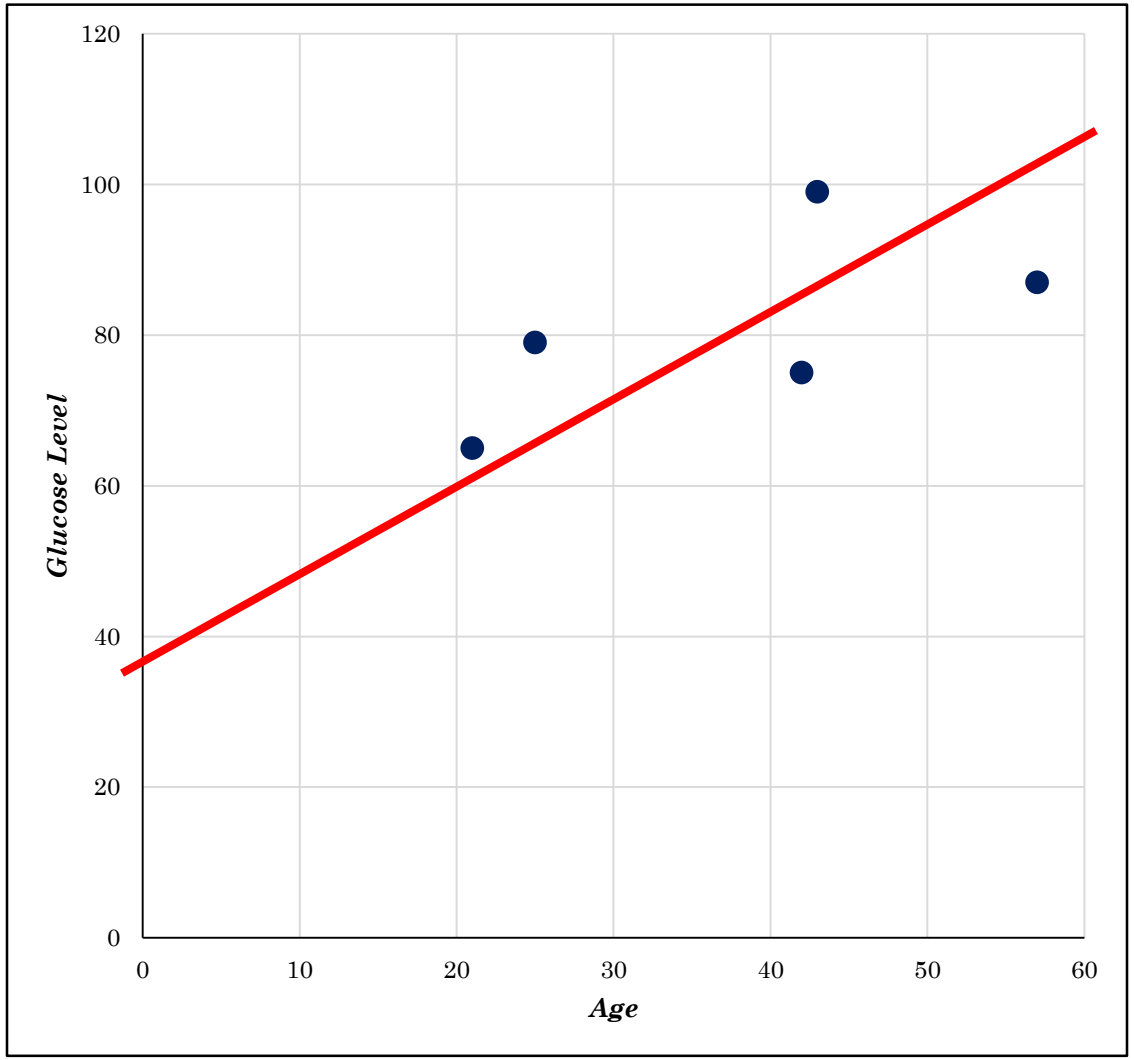
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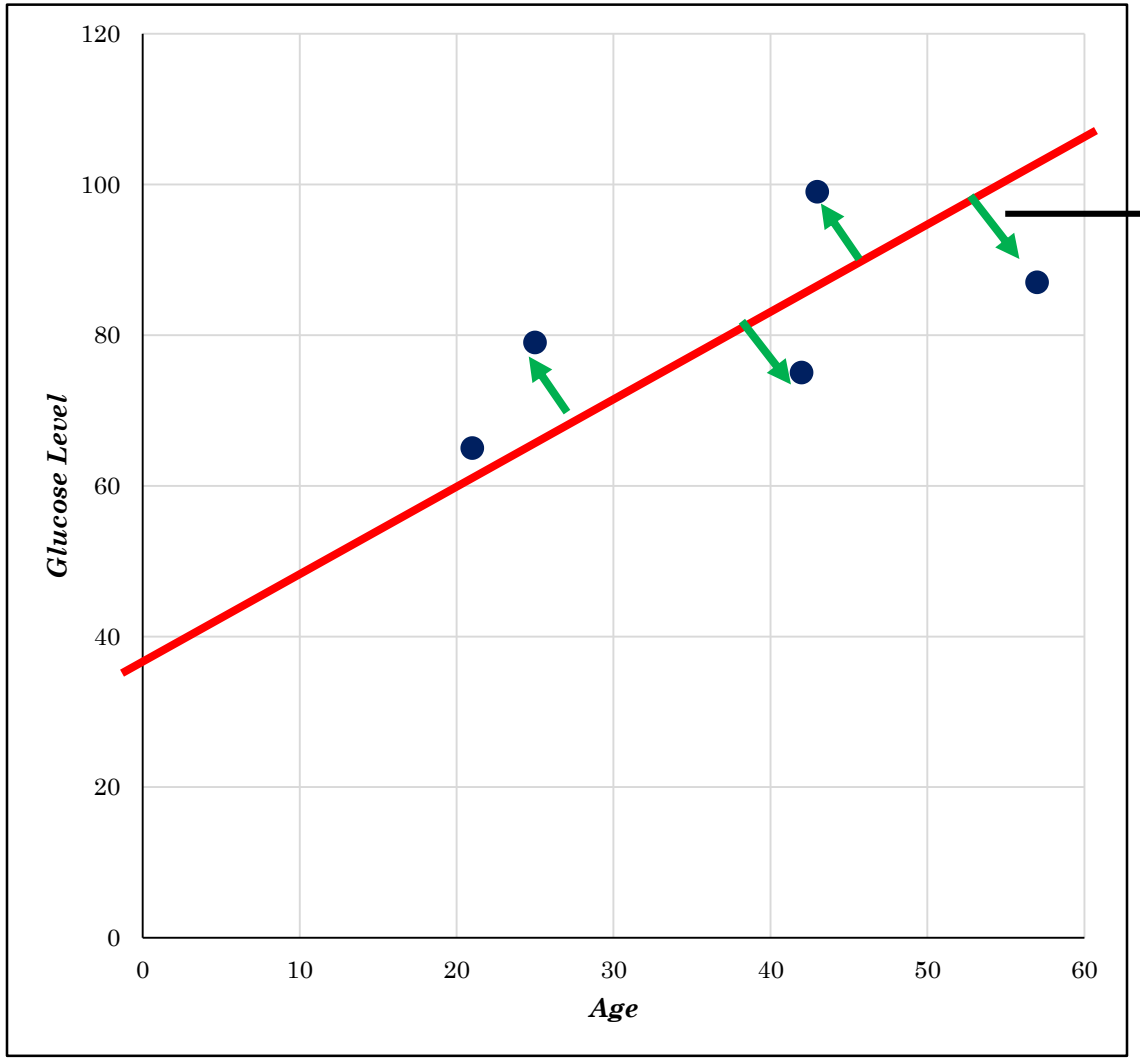
Estimating parameters



Estimating parameters



Estimating parameters



$$Y_i = \alpha + \beta X_i + \epsilon_i$$

Estimating parameters

- Let the estimator of α and β is $\hat{\alpha}$ and $\hat{\beta}$.
- $\hat{\beta} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$
- $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$

Estimating parameters

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- $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$

$$\widehat{Y_i} = \hat{\alpha} + \hat{\beta} X_i$$

Fitted model

Example

- Example: Following are the Advertising Expenditure and Sales (in 000$) Revenue of 10 months of a company.

Expenditure	**25**	**32**	**27**	**28**	**34**	**30**
Sales	**470**	**535**	**500**	**510**	**575**	**550**

a) **Draw the scatter diagram.**

b) Determine the fitted regression model of sales on advertising expenditure.

c) Interpret the regression parameter.

d) Estimate sales revenue when advertising expenditure is 35.

Example

$$\hat{\beta} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

x(Expenditure)	y (Sales)	x_i^2	$x_i \times y_i$
25	470	625	11750
32	535	1024	17120
27	500	729	13500
28	510	784	14280
34	575	1156	19550
30	550	900	16500
$\sum x_i = 176$	$\sum y_i = 3140$	$\sum x_i^2 = 5218$	$\sum x_i y_i = 92700$

Example

$$\hat{\beta} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$\hat{\beta} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} = \frac{(6\times 92700) - (176\times 3140)}{(6\times 5118) - (176)^2} = 10.72$$

$\hat{\beta} = 10.72$ means that an increase of \$1000 in advertising expenditure, the average sales revenue will increase \$10720.

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} = 523.34 - (10.72 \times 29.74) = ?$$

$\hat{\alpha} = ?$ means that, if there is no advertising expenditure then average sales revenue would be \$204530

$$Fitted\ model: \widehat{Y_i} = 204.53 + 10.72X_i$$

$$If\ X = 35: \widehat{Y_i} = 204.53 + 10.72 * 35 = 579.73$$

Goodness of Fit

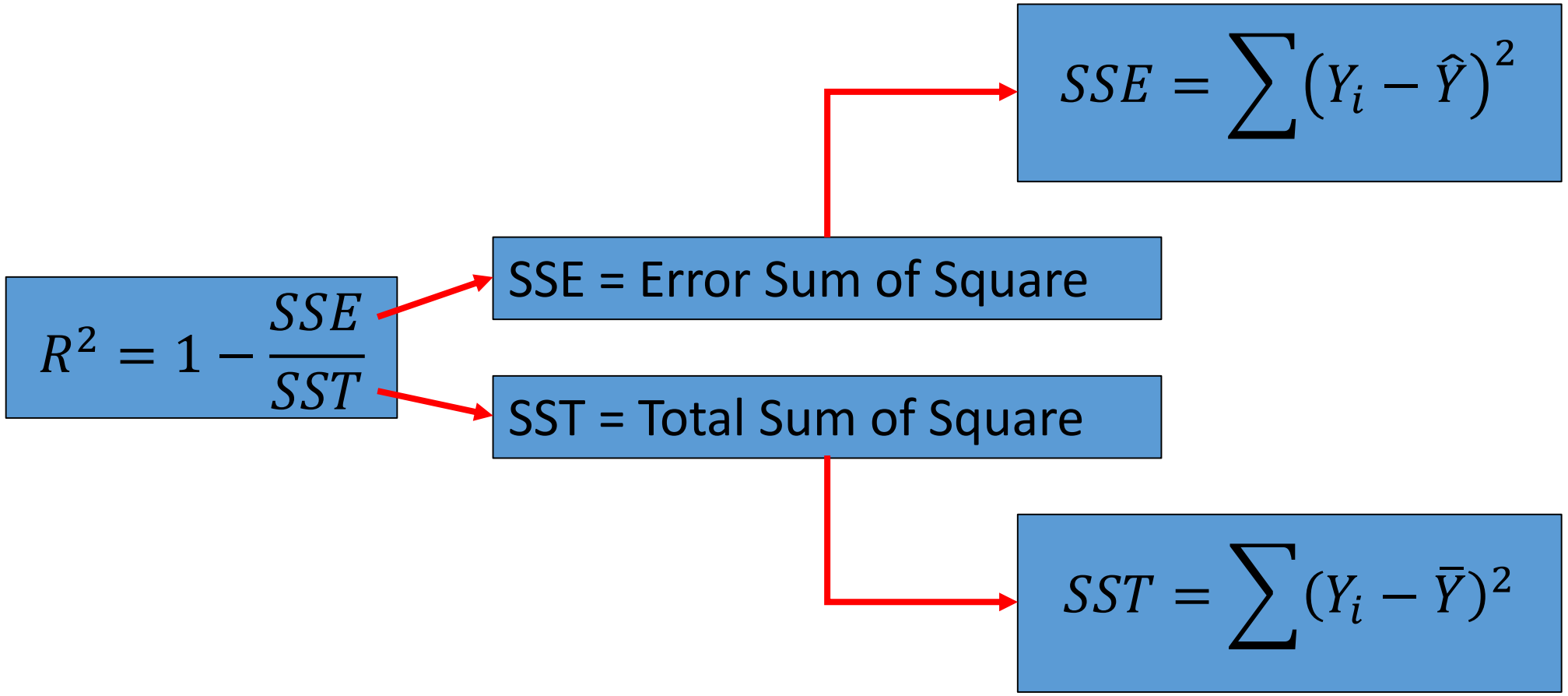
- How precise such predictions are?
- Is this regression equation is useful for prediction?
- To answer these questions, we need “Coefficient of Determination”- Denoted by $\boldsymbol{R^2}$

Goodness of Fit

- The coefficient of determination tells the percent of the variation in the dependent variable that is explained (determined) by the model and the explanatory variable. It is denoted by R^2

- **Range of R^2: [0 *to* 1]**
- $\boldsymbol{R^2 = 0}$: ***Equation is not useful for predictions***
- $\boldsymbol{R^2 = 1}$: ***Equation is useful for predictions***

Coefficient of determination



Coefficient of determination

$$\widehat{Y}_i = 204.53 + 10.72X_i$$

x(Expenditure)	y (Sales)	$\widehat{Y_i} = 204.53 + 10.72X_i$	$(y_i - \hat{y}_i)^2$	$(y_i - \bar{y})^2$
25	470	472.53	6.4	2845.15
32	535	547.57	158	135.56
27	500	493.97	36.36	544.76
28	510	504.69	28.20	177.73
34	575	569.01	5.99	2,668.75
30	550	526.13	569.78	710.76
$\sum x_i = 176$	$\sum y_i$ $= 3140$		$\sum (y_i - \hat{y}_i)^2$ $= 804.73$	$\sum (y_i - \bar{y})^2$ $= 7082.71$

$$SSE = \sum (Y_i - \hat{Y})^2$$

$$SST = \sum (Y_i - \bar{Y})^2$$

Coefficient of determination

$$R^2 = 1 - \frac{SSE}{SST} \Rightarrow R^2 = 1 - \frac{804.73}{7082.71} \Rightarrow R^2 = 0.89$$

Interpretation of R^2: For example, $R^2 = 0.89$ or 89%. It indicates that, almost 89% of the variability of the dependent variables explained by the independent variables.

Properties of Regression Coefficients

1. Regression coefficient has unit.

2. Regression coefficients are not symmetric function, $i.e., \beta_{yx} \neq \beta_{xy}$

Thank You